Football API

Overview

REST API for football match and team data with HATEOAS and JWT authentication

Version information

Version : 1.0

URI scheme

Host : localhost:8080
BasePath : /api/v1

Security

Bearer

Type "Bearer" followed by a space and JWT token.

Type : apiKey
Name : Authorization
In : HEADER

Paths

User login

```
POST /auth/login
```

Description

Authenticate user and return JWT token

Parameters

Type	Name	Description	Schema
Body	**request** *required*	User login credentials	models.LoginRequest

Responses

HTTP Code	Description	Schema
200	Login successful	models.LoginResponse
400	Invalid request	models.ErrorResponse
401	Invalid credentials	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Consumes

- `application/json`

Produces

- `application/json`

Tags

- auth

Register a new user

```
POST /auth/register
```

Description

Create a new user account with username and password

Parameters

Type	Name	Description	Schema
Body	**request** *required*	User registration details	models.RegisterRequest

Responses

HTTP Code	Description	Schema
201	User created successfully	object
400	Invalid request	models.ErrorResponse
409	Username already exists	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Consumes

- `application/json`

Produces

- `application/json`

Tags

- auth

Get head-to-head matches

```
GET /football/head-to-head
```

Description

Get all matches between two specific teams

Parameters

Type	Name	Description	Schema
Query	**teamA** *required*	First team ID	integer
Query	**teamB** *required*	Second team ID	integer

Responses

HTTP Code	Description	Schema
200	Head-to-head matches	models.MatchesResponse
400	Invalid query parameters	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- matches

Create a new match

```
POST /football/matches
```

Description

Create a new match result (requires authentication)

Parameters

Type	Name	Description	Schema
Body	**request** *required*	Match details	models.CreateMatchRequest

Responses

HTTP Code	Description	Schema
201	Match created	models.MatchResponse
400	Invalid request	models.ErrorResponse
401	Unauthorized	models.ErrorResponse

HTTP Code	Description	Schema
409	Match already exists	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Consumes

- `application/json`

Produces

- `application/json`

Tags

- matches

Security

Type	Name
apiKey	**Bearer**

List all matches

```
GET /football/matches
```

Description

Get all matches with pagination support

Parameters

Type	Name	Description	Schema	Default
Query	**limit** *optional*	Number of results per page	integer	`50`
Query	**offset** *optional*	Offset for pagination	integer	`0`

Responses

HTTP Code	Description	Schema
200	List of matches	models.MatchesResponse
400	Invalid query parameters	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- matches

Simulate a match outcome

```
POST /football/matches/simulate
```

Description

Runs a Monte Carlo simulation for a potential match result using historical Elo ratings and scoring patterns.

Parameters

Type	Name	Description	Schema
Body	**body** *required*	Simulation parameters	models.SimulateRequest

Responses

HTTP Code	Description	Schema
200	Simulation result	models.SimulateResponse
400	Invalid input	models.ErrorResponse
401	Unauthorized	models.ErrorResponse

HTTP Code	Description	Schema
429	Too many concurrent simulation requests	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Consumes

- `application/json`

Produces

- `application/json`

Tags

- matches

Security

Type	Name
apiKey	**Bearer**

Get a match by ID

```
GET /football/matches/{id}
```

Description

Get detailed information about a specific match

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Match ID	integer

Responses

HTTP Code	Description	Schema
200	Match details	models.MatchResponse
400	Invalid match ID	models.ErrorResponse
404	Match not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- matches

Update a match

```
PUT /football/matches/{id}
```

Description

Update an existing match record (requires authentication)

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Match ID	integer
Body	**request** *required*	Updated match details	models.UpdateMatchRequest

Responses

HTTP Code	Description	Schema
200	Match updated	models.MatchResponse

HTTP Code	Description	Schema
400	Invalid request	models.ErrorResponse
401	Unauthorized	models.ErrorResponse
404	Match not found	models.ErrorResponse
409	Match already exists	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Consumes

- `application/json`

Produces

- `application/json`

Tags

- matches

Security

Type	Name
apiKey	**Bearer**

Delete a match

```
DELETE /football/matches/{id}
```

Description

Delete a match by ID (requires authentication)

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Match ID	integer

Responses

HTTP Code	Description	Schema
204	Match deleted successfully	No Content
400	Invalid match ID	models.ErrorResponse
401	Unauthorized	models.ErrorResponse
404	Match not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- matches

Security

Type	Name
apiKey	**Bearer**

Create a goal

```
POST /football/matches/{id}/goals
```

Description

Record a new goal for a match

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Match ID	integer
Body	**goal** *required*	Goal details	models.CreateGoalRequest

Responses

HTTP Code	Description	Schema
201	Single created goal wrapped in a GoalsResponse collection	models.GoalsResponse
400	Invalid input	models.ErrorResponse
401	Unauthorized	models.ErrorResponse
404	Match or team not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Consumes

- `application/json`

Produces

- `application/json`

Tags

- goals

Security

Type	Name
apiKey	**Bearer**

Get goals for a match

```
GET /football/matches/{id}/goals
```

Description

Get all goals scored in the specified match

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Match ID	integer

Responses

HTTP Code	Description	Schema
200	List of goals	models.GoalsResponse
400	Invalid match ID	models.ErrorResponse
404	Match not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- goals

Delete a goal

```
DELETE /football/matches/{id}/goals/{goalId}
```

Description

Remove a goal record from a match

Parameters

Type	Name	Description	Schema
Path	**goalId** *required*	Goal ID	integer
Path	**id** *required*	Match ID	integer

Responses

HTTP Code	Description	Schema
204	Goal deleted	No Content
400	Invalid ID	models.ErrorResponse
401	Unauthorized	models.ErrorResponse
404	Goal not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Tags

- goals

Security

Type	Name
apiKey	**Bearer**

Create a shootout

```
POST /football/matches/{id}/shootout
```

Description

Record a penalty shootout result for a match

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Match ID	integer
Body	**shootout** *required*	Shootout details	models.CreateShootoutRequest

Responses

HTTP Code	Description	Schema
201	Shootout created	models.ShootoutResponse
400	Invalid input	models.ErrorResponse
401	Unauthorized	models.ErrorResponse
404	Match or team not found	models.ErrorResponse
409	Shootout already exists	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Consumes

- `application/json`

Produces

- `application/json`

Tags

- shootouts

Security

Type	Name
apiKey	**Bearer**

Get shootout result

```
GET /football/matches/{id}/shootout
```

Description

Get the penalty shootout result for a match

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Match ID	integer

Responses

HTTP Code	Description	Schema
200	Shootout result	models.ShootoutResponse
400	Invalid match ID	models.ErrorResponse
404	Match or shootout not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- shootouts

Delete a shootout

```
DELETE /football/matches/{id}/shootout
```

Description

Remove a penalty shootout record from a match

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Match ID	integer

Responses

HTTP Code	Description	Schema
204	Shootout deleted	No Content
400	Invalid match ID	models.ErrorResponse
401	Unauthorized	models.ErrorResponse
404	Shootout not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Tags

- shootouts

Security

Type	Name
apiKey	**Bearer**

Get goals by player

```
GET /football/players/{name}/goals
```

Description

Get all goals scored by a specific player

Parameters

Type	Name	Description	Schema
Path	**name** *required*	Player name	string

Responses

HTTP Code	Description	Schema
200	List of goals	models.GoalsResponse

HTTP Code	Description	Schema
400	Invalid player name	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- players

Get global Elo rankings

```
GET /football/rankings/elo
```

Description

Returns a paginated snapshot of global Elo rankings, optionally filtered by region

Parameters

Type	Name	Description	Schema
Query	**date** *optional*	Point-in-time date (YYYY-MM-DD)	string
Query	**limit** *optional*	Page size (default 50)	integer
Query	**offset** *optional*	Page offset (default 0)	integer
Query	**region** *optional*	Filter by region (e.g. europe, asia)	string

Responses

HTTP Code	Description	Schema
200	Global Elo rankings	elo.RankingsResponse

HTTP Code	Description	Schema
400	Invalid query parameters	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- elo

Recalculate Elo rankings

```
POST /football/rankings/elo/recalculate
```

Description

Triggers background Elo recalculation (admin). Optionally scoped to one team.

Parameters

Type	Name	Description	Schema
Query	**force** *optional*	Force recalculation even if cache is current	boolean
Query	**team_id** *optional*	Limit recalculation to this team	integer

Responses

HTTP Code	Description	Schema
202	Recalculation started	elo.RecalculateResponse
400	Invalid parameters	models.ErrorResponse
401	Unauthorized	models.ErrorResponse

HTTP Code	Description	Schema
429	Recalculation already running or rate limit exceeded	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- elo

Security

Type	Name
apiKey	**Bearer**

Create a new team

```
POST /football/teams
```

Description

Create a new national team (requires authentication)

Parameters

Type	Name	Description	Schema
Body	**request** *required*	Team details	models.CreateTeamRequest

Responses

HTTP Code	Description	Schema
201	Team created	models.TeamResponse
400	Invalid request	models.ErrorResponse

HTTP Code	Description	Schema
401	Unauthorized	models.ErrorResponse
409	Team already exists	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Consumes

- `application/json`

Produces

- `application/json`

Tags

- teams

Security

Type	Name
apiKey	**Bearer**

List all teams

```
GET /football/teams
```

Description

Get all national teams with HATEOAS links

Responses

HTTP Code	Description	Schema
200	List of teams	models.TeamsResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- teams

Get a team by ID

```
GET /football/teams/{id}
```

Description

Get detailed information about a specific team

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Team ID	integer

Responses

HTTP Code	Description	Schema
200	Team details	models.TeamResponse
400	Invalid team ID	models.ErrorResponse
404	Team not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- teams

Update a team

```
PUT /football/teams/{id}
```

Description

Update team name (requires authentication)

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Team ID	integer
Body	**request** *required*	Updated team details	models.UpdateTeamRequest

Responses

HTTP Code	Description	Schema
200	Team updated	models.TeamResponse
400	Invalid request	models.ErrorResponse
401	Unauthorized	models.ErrorResponse
404	Team not found	models.ErrorResponse
409	Team name already in use	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Consumes

- `application/json`

Produces

- `application/json`

Tags

- teams

Security

Type	Name
apiKey	**Bearer**

Delete a team

```
DELETE /football/teams/{id}
```

Description

Delete a team by ID (requires authentication)

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Team ID	integer

Responses

HTTP Code	Description	Schema
204	Team deleted successfully	No Content
400	Invalid team ID	models.ErrorResponse
401	Unauthorized	models.ErrorResponse
404	Team not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- teams

Security

Type	Name
apiKey	**Bearer**

Get team Elo rating

```
GET /football/teams/{id}/elo
```

Description

Returns the World Football Elo rating for a team, optionally at a historical date

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Team ID	integer
Query	**date** *optional*	Point-in-time date (YYYY-MM-DD); defaults to today	string
Query	**include_history** *optional*	Include full rating history	boolean

Responses

HTTP Code	Description	Schema
200	Team Elo rating	elo.Rating
400	Invalid team ID or date	models.ErrorResponse
404	Team not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- elo

Get team Elo timeline

```
GET /football/teams/{id}/elo/timeline
```

Description

Returns the full time-series of Elo rating changes for a team

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Team ID	integer
Query	**end_date** *optional*	End date (YYYY-MM-DD)	string
Query	**resolution** *optional*	Aggregation: match|month|year (default: match)	string
Query	**start_date** *optional*	Start date (YYYY-MM-DD)	string

Responses

HTTP Code	Description	Schema
200	Team Elo timeline	elo.TimelineResponse
400	Invalid team ID or date	models.ErrorResponse
404	Team not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- elo

Get team history

```
GET /football/teams/{id}/history
```

Description

Get historical names for a specific team

Parameters

Type	Name	Description	Schema
Path	**id** *required*	Team ID	integer

Responses

HTTP Code	Description	Schema
200	Team history	models.FormerNamesResponse
400	Invalid team ID	models.ErrorResponse
404	Team not found	models.ErrorResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- teams

List all tournaments

```
GET /football/tournaments
```

Description

Get all football tournaments ordered by name

Responses

HTTP Code	Description	Schema
200	List of tournaments	models.TournamentsResponse
500	Internal server error	models.ErrorResponse

Produces

- `application/json`

Tags

- tournaments

Definitions

elo.Methodology

Name	Schema
formulaReference *optional*	string
homeAdvantage *optional*	number
kFactor *optional*	number
weightMultiplier *optional*	number

elo.RankingEntry

Name	Schema
elo *optional*	number
links *optional*	< models.Link > array
rank *optional*	integer
teamId *optional*	integer
teamName *optional*	string

elo.RankingsResponse

Name	Schema
data *optional*	< elo.RankingEntry > array
date *optional*	string
limit *optional*	integer
links *optional*	< models.Link > array
offset *optional*	integer
total *optional*	integer

elo.Rating

Name	Schema
changeFromPrevious *optional*	number
date *optional*	string
elo *optional*	number
links *optional*	< models.Link > array

Name	Schema
matchesConsidered *optional*	integer
methodology *optional*	elo.Methodology
rank *optional*	integer
teamId *optional*	integer
teamName *optional*	string

elo.RecalculateResponse

Name	Schema
links *optional*	< models.Link > array
message *optional*	string

elo.TimelineEntry

Name	Description	Schema
change *optional*		number
date *optional*		string
elo *optional*		number
homeAway *optional*	"H", "A", or "N"	string
matchId *optional*		integer
opponent *optional*		string
result *optional*	"W", "D", or "L"	string

elo.TimelineResponse

Name	Schema
data *optional*	< elo.TimelineEntry > array
links *optional*	< models.Link > array
teamId *optional*	integer
teamName *optional*	string

models.CreateGoalRequest

Name	Description	Schema
ownGoal *optional*		boolean
penalty *optional*		boolean
scorer *required*	**Length** : `1 - 100`	string
teamId *required*	**Minimum value** : `1`	integer

models.CreateMatchRequest

Name	Description	Schema
awayScore *optional*	**Minimum value** : `0`	integer
awayTeamId *required*	**Minimum value** : `1`	integer
city *optional*		string
country *optional*		string
date *required*		string
homeScore *optional*	**Minimum value** : `0`	integer

Name	Description	Schema
homeTeamId *required*	**Minimum value** : 1	integer
neutral *optional*		boolean
tournamentId *required*	**Minimum value** : 1	integer

models.CreateShootoutRequest

Name	Description	Schema
winnerId *required*	**Minimum value** : 1	integer

models.CreateTeamRequest

Name	Description	Schema
name *required*	**Length** : `1 - 100`	string

models.ErrorResponse

Name	Schema
error *optional*	string

models.ExpectedScore

Name	Schema
awayGoals *optional*	number
homeGoals *optional*	number

models.FormerName

Name	Schema
endDate *optional*	string

Name	Schema
formerName *optional*	string
id *optional*	integer
startDate *optional*	string
teamId *optional*	integer

models.FormerNamesResponse

Name	Schema
data *optional*	< models.FormerName > array
links *optional*	< models.Link > array

models.Goal

Name	Schema
id *optional*	integer
matchId *optional*	integer
ownGoal *optional*	boolean
penalty *optional*	boolean
scorer *optional*	string
team *optional*	string
teamId *optional*	integer

models.GoalsResponse

Name	Schema
data *optional*	< models.Goal > array
links *optional*	< models.Link > array

models.Link

Name	Schema
href *optional*	string
method *optional*	string
rel *optional*	string

models.LoginRequest

Name	Schema
password *required*	string
username *required*	string

models.LoginResponse

Name	Schema
links *optional*	< models.Link > array
token *optional*	string

models.MatchResponse

Name	Schema
awayScore *optional*	integer
awayTeam *optional*	string

Name	Schema
awayTeamId *optional*	integer
city *optional*	string
country *optional*	string
date *optional*	string
homeScore *optional*	integer
homeTeam *optional*	string
homeTeamId *optional*	integer
id *optional*	integer
links *optional*	< models.Link > array
neutral *optional*	boolean
tournament *optional*	string
tournamentId *optional*	integer

models.MatchesResponse

Name	Schema
data *optional*	< models.MatchResponse > array
links *optional*	< models.Link > array

models.RegisterRequest

Name	Description	Schema
password *required*	**Length** : 8 - 128	string

Name	Description	Schema
username *required*	**Length** : `3 - 50`	string

models.ShootoutResponse

Name	Schema
id *optional*	integer
links *optional*	< models.Link > array
matchId *optional*	integer
winner *optional*	string
winnerId *optional*	integer

models.SimulateRequest

Name	Description	Schema
awayTeamId *required*	AwayTeamID is the ID of the team designated as "away" for the simulation. **Minimum value** : `1`	integer
date *optional*	Date is an optional point-in-time date (YYYY-MM-DD) used to derive Elo ratings from historical data. Defaults to today when omitted.	string
homeTeamId *required*	HomeTeamID is the ID of the team designated as "home" for the simulation. **Minimum value** : `1`	integer
simulations *optional*	Simulations is the number of Monte Carlo iterations to run. Values ≤ 0 default to 1 000; values > 10 000 are capped at 10 000.	integer
venue *optional*	Venue describes where the match is to be played: "home" (home team's ground), "away" (away team's ground / home team away), or "neutral". Defaults to "neutral" when omitted or invalid.	string

models.SimulateResponse

Name	Schema
asOf *optional*	string
awayElo *optional*	number
awayTeam *optional*	string
expectedScore *optional*	models.ExpectedScore
homeElo *optional*	number
homeTeam *optional*	string
links *optional*	< models.Link > array
outcome *optional*	models.SimulationOutcome
simulations *optional*	integer
upsetProbability *optional*	number
venue *optional*	string

models.SimulationOutcome

Name	Schema
awayWinCI *optional*	< number > array
awayWinPct *optional*	number
drawCI *optional*	< number > array
drawPct *optional*	number
homeWinCI *optional*	< number > array
homeWinPct *optional*	number

models.TeamResponse

Name	Schema
createdAt *optional*	string
id *optional*	integer
links *optional*	< models.Link > array
name *optional*	string

models.TeamsResponse

Name	Schema
data *optional*	< models.TeamResponse > array
links *optional*	< models.Link > array

models.Tournament

Name	Schema
createdAt *optional*	string
id *optional*	integer
name *optional*	string

models.TournamentsResponse

Name	Schema
data *optional*	< models.Tournament > array

models.UpdateMatchRequest

Name	Description	Schema
awayScore *optional*	**Minimum value** : `0`	integer
awayTeamId *required*	**Minimum value** : `1`	integer
city *optional*		string
country *optional*		string
date *required*		string
homeScore *optional*	**Minimum value** : `0`	integer
homeTeamId *required*	**Minimum value** : `1`	integer
neutral *optional*		boolean
tournamentId *required*	**Minimum value** : `1`	integer

models.UpdateTeamRequest

Name	Description	Schema
name *required*	**Length** : `1 - 100`	string